

Circuit and working

Normally, in an elevator system, when the carriage arrives at a landing, it throws a crowbar, which, in turn, actuates a number of switches. These switches may be on the +24V side or on the ground side. In our case, these are on the ground side, which means the +24V indication lamps have the +24V side already connected and ready. And when the switch actuates, it sends the ground connection to the bulb to switch it on.

Similarly, when the carriage moves past that landing, the crowbar releases the switch along with the ground connections, and the bulb goes off. However, for some older lifts, these signals come from the +24V side rather than the ground. These are then inverted using a simple transistor (BC547) switching technique as shown in the schematic (see Floor N). After inverting these, these can be used again, as shown in the schematic.

We pick up ground connections (using SW1, SW2 and SW3) for our Raspi based indications. All these connections reach inside the carriage as well as outside the landings. Inside and outside indication is a series of horizontal +24V bulbs placed under a bar, indicating the floor numbers. When the lift cage reaches a floor, the bulb under that floor number glows.

To have different addresses on I2C bus of MCP23008 and MCP23017, we have connected the address ports in different ways to have 0x20 and 0x21 as addresses of these two different MCPs. Python script runs at boot level and monitors the GPIOs of MCP23017. The moment one landing

Important notes

Important note 1. The software is built on Raspi version 1 board. For running it on version 2 board, we need to make some changes in the following files:

```
lcd23008.py
Adafruit_MCP230xx.py
lift_light.py
lift-light-with-espeak.py
```

Where it needs to be changed, it is marked clearly like:

```
# use busnum=0 for Raspi version 1
(256MB) and busnum=1 for Raspi version 2
```

Important note 2. In case you are using the latest Raspbian OS, add the following lines:

```
in /boot/config.txt & reboot
```

```
$sudo nano /boot/config.txt
```

```
dtparam=spi=on
dtparam=i2c_arm=on
dtoverlay=w1-gpio
```

Save nano by ctrl+o and then exit by ctrl+x.

Make the changes accordingly and then reboot to make the effects permanent.

arrives, it shorts the concerned GPIO with the ground, thereby switching the respective MP3 file and the LCD panel showing the respective floor number, besides the clock and temperature signals. Temperature sensor DS18B20 is used for sensing lift-cage temperature.

Program

The program is built using easy Python scripts. Additional software that we have used are all free Adafruit libraries for manipulating the GPIOs of MCP230xx, one Python GPIO for manipulating the software and one mpg321 for playing MP3 files. Ada-

fruit libraries may be copied from my repository here.

<https://drive.google.com/file/d/0B3E3LcSKoM-6S21ZUnZOd-WZJRGs/edit?usp=sharing>

However, mpg321 and GPIOs have to be installed from the Internet.

mpg321 may be installed using the following command:

```
$> sudo apt-get install mpg321
```

Operation

MP3 files are to be made separately from recordings using anorak or some other recording program. Each MP3 will speak about different floors, namely, first floor, second floor, third floor and so on.

A small speaker amplifier is added to the audio-out port of Raspi. Power supply of the audio amplifier is taken separately (normally, 6V/12V). To run the program at boot level, so that every time Raspi restarts after any power failure, add the following line at /etc/rc.local file:

```
$> sudo nano /etc/rc.local
```

#add this line before exit command:

```
sudo python lift_light.py &
```

Save nano with ctrl+o and then exit with ctrl+x

To run the program from the command line, use:

```
$> sudo python lift_light.py &
```

Alternate software

To get rid of MP3 files, we can train our Raspi to use a synthesiser program to speak up. There are many voice synthesiser software available online, but eSpeak is the most versatile and it comes free of cost. The other voice synthesiser software are



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